Script Title

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Chapter 1

Limits

Definition 1.1 Limit

Let $f : D \longrightarrow \mathbb{R}$ be a function with $D \subseteq \mathbb{R}$ and a a cluster point of D. Then L is the limit of f at the point a, if for all $\varepsilon > 0$ there exists a $\delta > 0$, such that for all $x \in D$ with $0 < |x - a| < \delta$ we have: $|f(x) - L| < \varepsilon$. We write:

$$\lim_{x \longrightarrow a} f(x) = L.$$

Example 1.2

Consider the function $f(x) = 2x + 3$. We want to determine the limit of f at the point $a = 1$. We suspect that $\lim_{x \to 1} f(x) = 5$, since $f(1) = 2 \cdot 1 + 3 = 5$.

To show this, we take an arbitrary $\varepsilon > 0$ and need to find a $\delta > 0$, such that for all x with $0 < |x - 1| < \delta$ we have: $|f(x) - 5| < \varepsilon$.

Let's calculate $|f(x) - 5|$:

$$|f(x) - 5| = |(2x + 3) - 5| = |2x - 2| = 2|x - 1|.$$

To ensure that $|f(x) - 5| < \varepsilon$, we set $2|x - 1| < \varepsilon$, which is equivalent to $|x - 1| < \frac{\varepsilon}{2}$.

Therefore, we can choose $\delta = \frac{\varepsilon}{2}$. Then for all x with $0 < |x - 1| < \delta$ we have:

$$|f(x) - 5| = 2|x - 1| < \varepsilon.$$

Thus, we have shown that $\lim_{x \to 1} f(x) = 5$.

Exercise 1.3

Calculate the limit $\lim_{x \to 0} \frac{\sin(x)}{x}$.

Proposition 1.4

Let $f : D \longrightarrow \mathbb{R}$ be a function with $D \subseteq \mathbb{R}$ and a a cluster point of D. If $\lim_{x \to a} f(x) = L$

exists, then this limit is unique.

Proof. Assume that there exist two different limits L_1 and L_2 of f at the point a. This means that for all $\varepsilon > 0$ there exists a $\delta_1 > 0$, such that for all $x \in D$ with $0 < |x - a| < \delta_1$ we have: $|f(x) - L_1| < \varepsilon$. Similarly, for all $\varepsilon > 0$ there exists a $\delta_2 > 0$, such that for all $x \in D$ with $0 < |x - a| < \delta_2$ we have: $|f(x) - L_2| < \varepsilon$.

Now choose $\varepsilon = \frac{|L_1 - L_2|}{3}$. Then there exist $\delta_1 > 0$ and $\delta_2 > 0$, such that for all x with $0 < |x - a| < \delta_1$ we have: $|f(x) - L_1| < \varepsilon$, and for all x with $0 < |x - a| < \delta_2$ we have: $|f(x) - L_2| < \varepsilon$. □

Corollary 1.5

If $f : D \to \mathbb{R}$ is a function with $D \subseteq \mathbb{R}$ and a a cluster point of D, and if $\lim_{x\to a} f(x) = L$ exists, then we also have $\lim_{x\to a} |f(x)| = |L|$.

Proof. Since $\lim_{x\to a} f(x) = L$ exists, for all $\varepsilon > 0$ there exists a $\delta > 0$, such that for all $x \in D$ with $0 < |x - a| < \delta$ we have: $|f(x) - L| < \varepsilon$. We want to show that $\lim_{x\to a} |f(x)| = |L|$.

Let's calculate $||f(x)| - |L||$:

$$||f(x)| - |L|| \leq |f(x) - L| < \varepsilon.$$

Therefore, for all x with $0 < |x - a| < \delta$ we have: $||f(x)| - |L|| < \varepsilon$. Thus, we have shown that $\lim_{x\to a} |f(x)| = |L|$. □

Fact 1.6

If $f : D \to \mathbb{R}$ is a function with $D \subseteq \mathbb{R}$ and a a cluster point of D, and if $\lim_{x\to a} f(x) = L$ exists, then we also have $\lim_{x\to a}(f(x) + c) = L + c$ for any constant $c \in \mathbb{R}$.

Theorem 1.7

Let $f : D \to \mathbb{R}$ be a function with $D \subseteq \mathbb{R}$ and a a cluster point of D. If $\lim_{x\to a} f(x) = L$ exists, then we also have $\lim_{x\to a}(c \cdot f(x)) = c \cdot L$ for any constant $c \in \mathbb{R}$.

Proof. Since $\lim_{x\to a} f(x) = L$ exists, for all $\varepsilon > 0$ there exists a $\delta > 0$, such that for all $x \in D$ with $0 < |x - a| < \delta$ we have: $|f(x) - L| < \varepsilon$. We want to show that $\lim_{x\to a}(c \cdot f(x)) = c \cdot L$.

Let's calculate $|c \cdot f(x) - c \cdot L|$:

$$|c \cdot f(x) - c \cdot L| = |c| \cdot |f(x) - L| < |c| \cdot \varepsilon.$$

□

Chapter 2

Continuity

Definition 2.1 Continuity

A function $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$ is continuous at the point $a \in D$, if

$$\lim_{x \to a} f(x) = f(a).$$

A function is continuous if it is continuous at every point in its domain.

Bibliography